University of Amsterdam
Amsterdam Business School

Data Science in Online Marketing

Online Advertising - Reinforcement Learning

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Course Outline

Date	Topic	Assignment
3 april	Recommenders: Content Based	
10 april	Recommenders: Collaborative Filtering	Yes
17 april	Recommenders: Guest Lecture Bol.com Online advertising: Introduction	
24 april	**Online advertising: Reinforcement Learning**	
1 may	Online advertising: CTR prediction	Yes
15 may	Online advertising: Media Mix Modeling & Conversion Attribution	
22 may	Revision	
29 may	Digital exam on location	

Measuring Performance in online advertising

Click-through rate (CTR) =

$$\frac{\#\ click\ throughs}{\#\ impressions}$$

Conversion rate (CR) =

$$\frac{conversions}{\#\ click\ throughs}$$

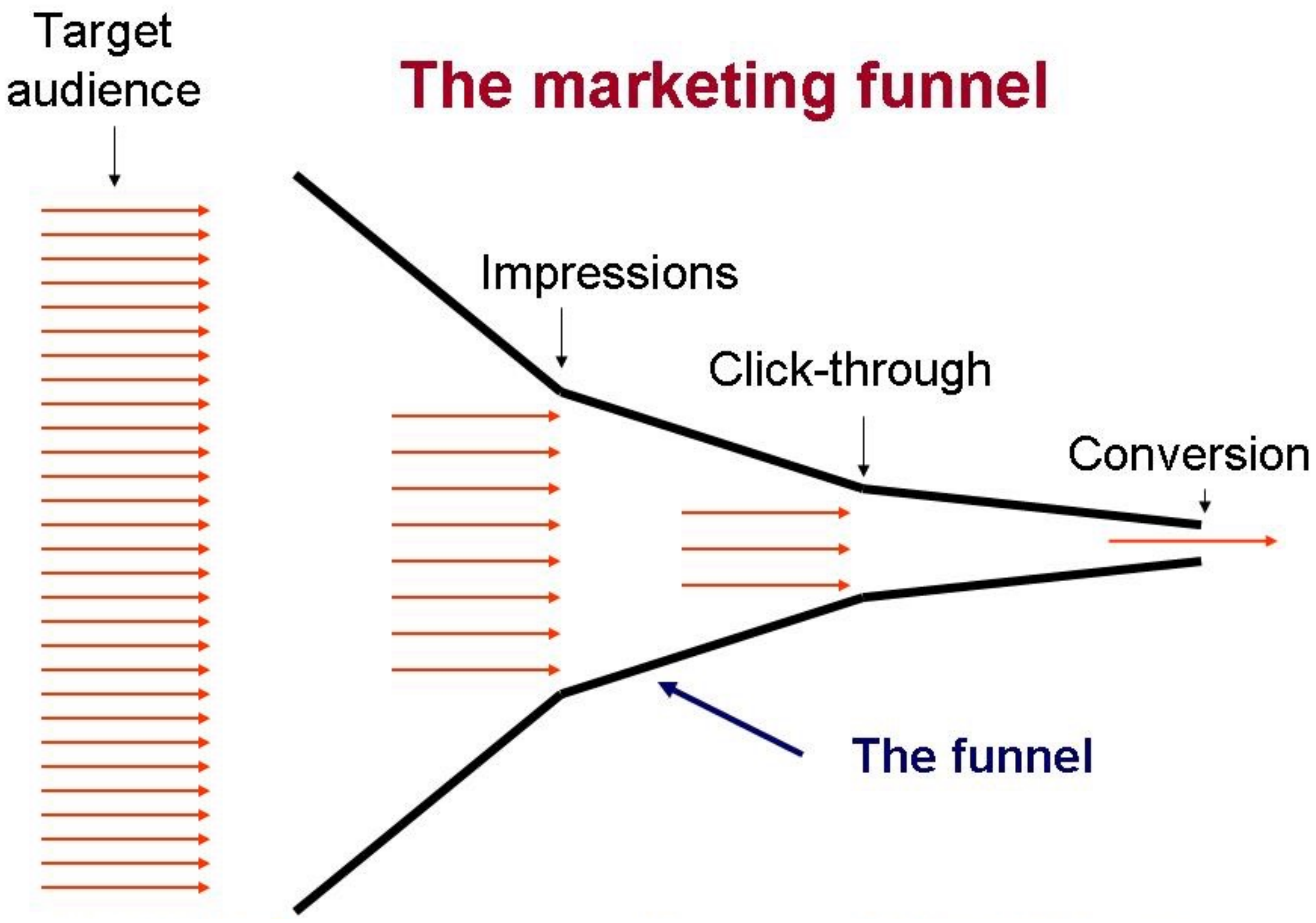

Image taken from: Customer Acquisition Cost, ibiz.wikidot.com

Click/Conversion Rate Optimization

- Test different versions of website/ad layout (**Explore**)
- Measure CTR / CR of different versions
- Use the version with the best performance (**Exploit**)

A/B Testing

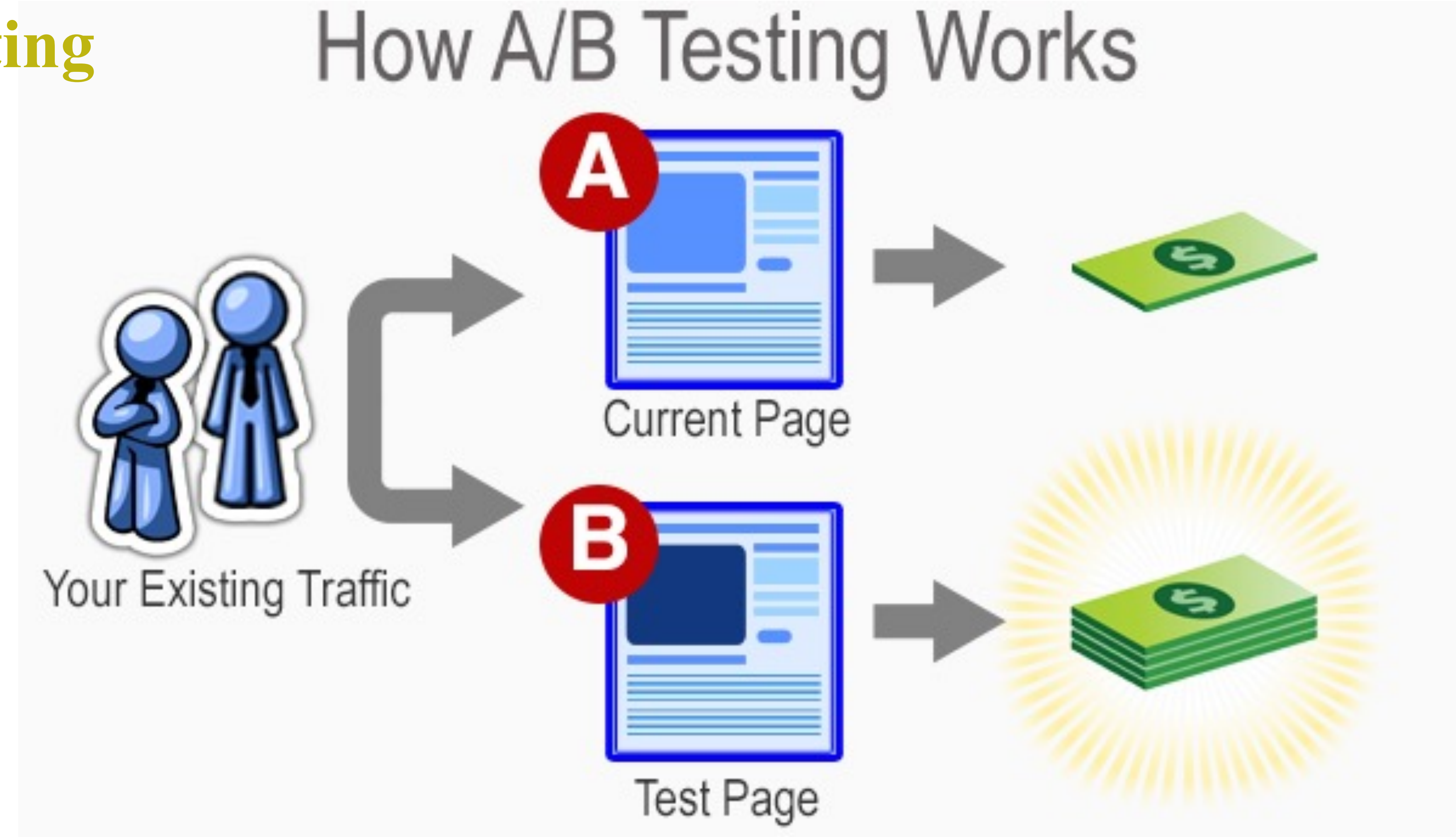

Image taken from: https://tapatihira.wordpress.com/2015/05/09/18/

A/B Testing

Question

- How much impressions do we need to measure CTR difference accurately?

Accuracy of Estimating CTR (or CR)

- Every impression leads to a click or not (1/0)
- Bernoulli experiment with $X_1, X_2, \ldots, X_n$
- Assume that version A has a CTR of 1% and version B of 1.1%.
- How many views n do we need to be 95% certain to detect that uplift?

$X_i \sim$ Bernoulli Distribution

$E(X_i) = p$
$\sigma(X_i) = \sqrt{p(1-p)}$

- CTR estimate: $Y_n = \sum_{i=1}^{n} X_i / n$ with $E(Y_n) = p$ and $\sigma(Y_n) = \sqrt{p(1-p)/n}$
- If we use Normal approximations and a Z test. $Z = \frac{p_b - p_a}{\sqrt{[(p_a(1-p_a)+(p_b(1-p_b)]/n}}$
- $Z \geq 1.95$ gives us 95% confidence: $\frac{p_b - p_a}{\sqrt{[(p_a(1-p_a)+(p_b(1-p_b)]/n}} \geq 1.95$
- $n \geq 80{,}000$ impressions per version

Accuracy of Estimating CTR

- We need many advertising impressions for a reliable CTR estimate
- But advertising costs money

How can we avoid wasting a lot of money before we have “sufficient” observations?

Bayesian Updating

Bayesian updating: “Learning on the job”

Classic statistics

1. Observe data
2. Estimate distribution

Bayesian inference

1. Assume distribution
2. Update distribution as we observe data

Bayesian updating: Simple example

Data:

	Head	Tail
Coin 1	2	0
Coin 2	45	5

Which coin would you play next if you bet on heads?

Bayesian updating: Simple example

Data:

	Head	Tail
Coin 1	2	0
Coin 2	45	5

Classic statistics

Estimate from data:

- Coin 1: 100% heads
- Coin 2: 90% heads

➢ Coin 1 has a better chance of heads
(not significant)

Bayesian updating: Simple example

Prior:

	Head	Tail
Coin 1	1	1
Coin 2	1	1

Posterior:

	Head	Tail
Coin 1	3	1
Coin 2	46	6

Data:

	Head	Tail
Coin 1	2	0
Coin 2	45	5

<u>Classic statistics</u>

Estimate from data:

- Coin 1: 100% heads
- Coin 2: 90% heads

➢ Coin 1 has a better chance of heads (not significant)

<u>Bayesian inference</u>

1. Assume prior distribution: both get 1 head and 1 tail (50% head probability)
2. Update distribution as we observe data:
   - Coin 1: 75% heads
   - Coin 2: 88% heads

➢ Coin 2 has a better chance of heads, look at distribution

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Convergence

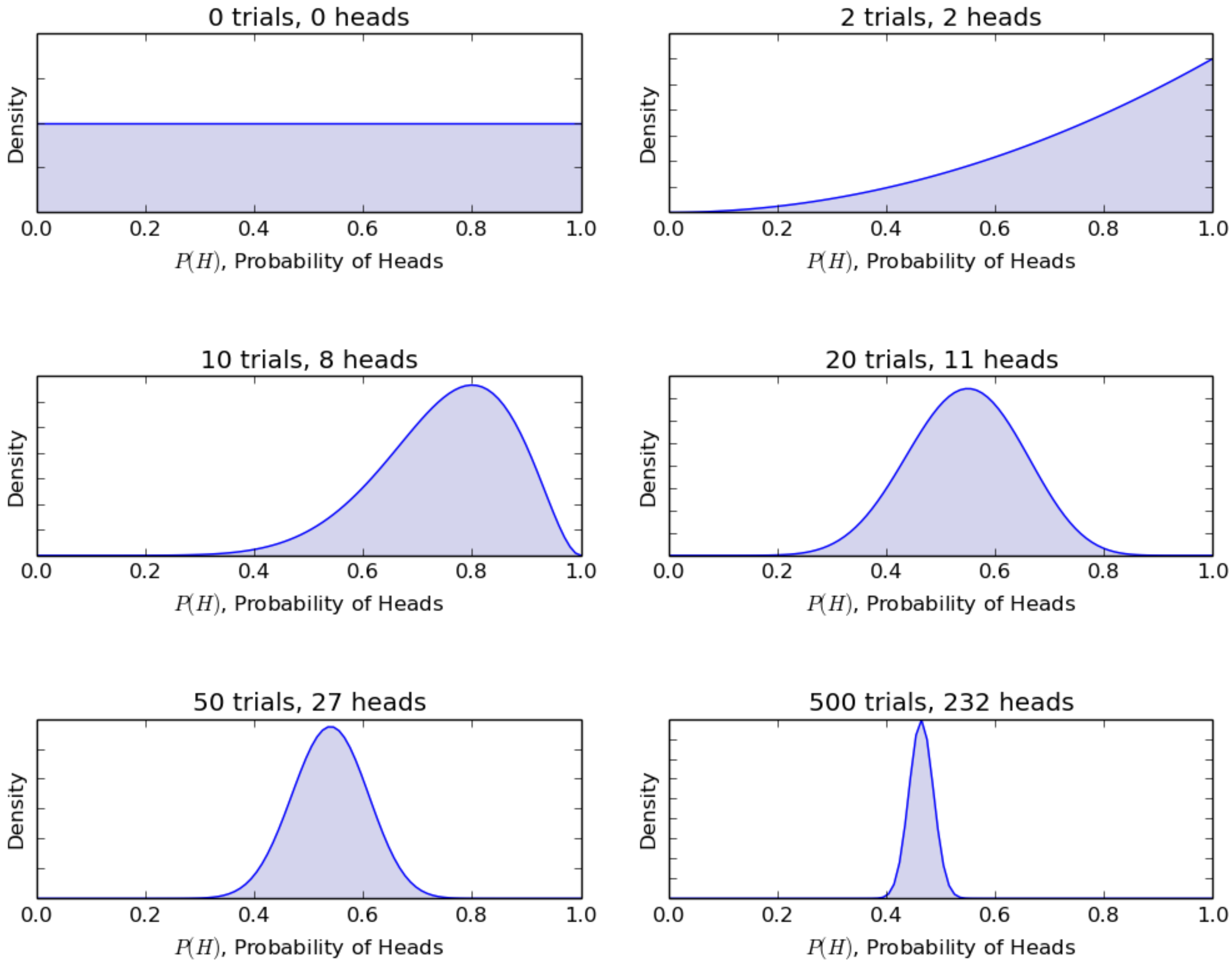


Image taken from: Bayesian Statistics Explained to Beginners in Simple Emglish, NSS, analyticsvidhya.com, June 2016

Challenges

- **How to choose the prior distribution?**
- **How to update as data comes in?**

- How to make your decision?

Bayes' Rule

Definition Conditional Probability:
$P(A|B) = P(A \cap B) / P(B)$

$$P(A|B) = P(B|A) \times \frac{P(A)}{P(B)}$$

P(A): (prior) Distribution function for the probability of heads: $f(x)$

B: result of last toss: Head or Tail

We want to know P(A|B) : (updated) distribution function for the probability of heads, given the last toss

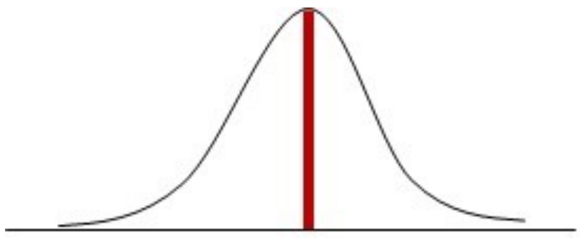
P(B|A) is simple to express in terms of x

P(B) is not directly known but can be calculated by $P(B) = \int_x P(B|A = x)P(A = x)$

Bayes' Rule at work

We want to estimate

$f(x)$ distribution function of the click rate probability x



$f_{prior}(x)$ prior distribution: estimate of distribution before observation

E Latest observervation ("evidence"): click or no-click

$f_{posterior}(x)$: posterior distribution: estimate of distribution after observation

$$f_{posterior}(x|E) = \mathrm{P}(E|x) \times \frac{f_{prior}(x)}{\mathrm{P}(E)} = \mathrm{P}(E|x) \times \frac{f_{prior}(x)}{\int_0^1 \mathrm{P}(E|z) \times f_{prior}(z)\, dz}$$

Bayes'Rule at work: updating

Each impression is a **Bernoulli experiment:**

With probility *p* we will observe a click and with probability *(1-p)* a no-click

$f_{prior}(x)$ prior distribution for p

E Latest observervation ("evidence"): click or no-click

$f_{posterior}(x)$: posterior distribution: estimate of distribution after observation

Note that: $\mathrm{P}(click|x) = x$ and $\mathrm{P}(no\ click|x) = 1 - x$

$$f_{posterior}(x|E) = \mathrm{P}(E|x) \times \frac{f_{prior}(x)}{\mathrm{P}(E)} = \mathrm{P}(E|x) \times \frac{f_{prior}(x)}{\int_0^1 \mathrm{P}(E|z) \times f_{prior}(z)\, dz}$$

Bayes'Rule at work: updating

Each impression is a **Bernoulli experiment:**

With probility *p* we will observe a click and with probability *(1-p)* a no-click

$f_{prior}(x)$ prior distribution for p

$f_{posterior}(x)$: posterior distribution: estimate of distribution after observation

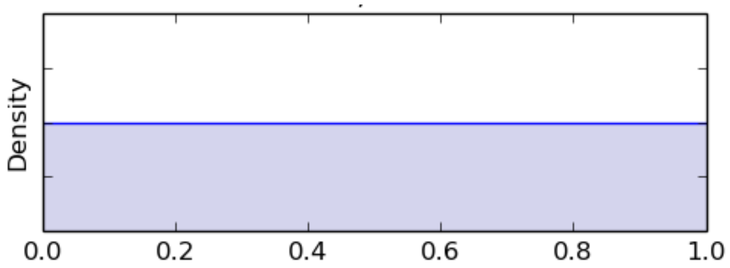
Note that: $\mathrm{P}(click|x) = x$ and $\mathrm{P}(no\ click|x) = 1 - x$

When we observe a click:

$$f_{posterior}(x|click) = \mathrm{P}(click|x) \times \frac{f_{prior}(x)}{\int_0^1 \mathrm{P}(click|z) \times f_{prior}(z)\, dz} = x \; \frac{f_{prior}(x)}{\int_0^1 z\, f_{prior}(z)\, dz}$$

Bayes'Rule at work: Uniform distribution as prior

Let's assume p has a **uniform(0,1) distribution** as prior



$f_{prior}(x) = 1$ prior distribution for p (click rate)

$\mathbb{E}_{prior}(x) = \frac{1}{2}$ Expected value of click rate for the prior distribution

$f_{posterior}(x)$: posterior distribution: estimate of distribution after observation

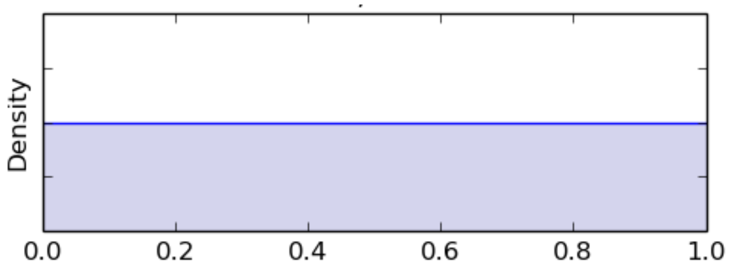
When we observe a click:

$f_{posterior}(x|click) =$

$\mathbb{E}_{posterior}(x) =$

Bayes'Rule at work: Uniform distribution as prior

Let's assume p has a **uniform(0,1) distribution** as prior



$f_{prior}(x) = 1$ prior distribution for p (click rate)

$\mathbb{E}_{prior}(x) = \frac{1}{2}$ Expected value of click rate for the prior distribution

$f_{posterior}(x)$: posterior distribution: estimate of distribution after observation

When we observe a click:

$$f_{posterior}(x|click) = x \ \frac{f_{prior}(x)}{\int_0^1 \ z\, f_{prior}(z)\, dz}) = \frac{x}{\int_0^1 \ z\, dz} = \frac{x}{\frac{1}{2}} = 2x$$

$$\mathbb{E}_{posterior}(x) = \int_0^1 2x^2 dx = \frac{2}{3}$$

Choosing a Prior Distribution

- **Uniform(0,1) distribution** is not a good prior in practice for CTR becasue every value has equal probability
- In practice often a low CTR ($<10\%$) is much more likely than a high CTR

Choosing a Prior Distribution: Beta Distribution

- Beta distribution has 2 parameters α,β: Beta(α,β)
- Prior distribution: $f_{prior}(x) = c\, x^{\alpha-1}\,(1-x)^{\beta-1}$
- $E_{prior}\,(CTR) = \frac{\alpha}{\alpha+\beta}$

▪ Beta prior results in a Beta Posterior!
▪ Interpretation of parameters:
  ▪ α: number of successes (clicks)
  ▪ β: number of failures (no-clicks)

After a view **with** click:

Posterior distribution: $f_{posterior} \sim \text{Beta}(\alpha+1,\beta)$

After a view **without** click:

Posterior distribution: $f_{posterior} \sim \text{Beta}(\alpha,\beta+1)$

Choosing a Prior Distribution: Beta Distribution

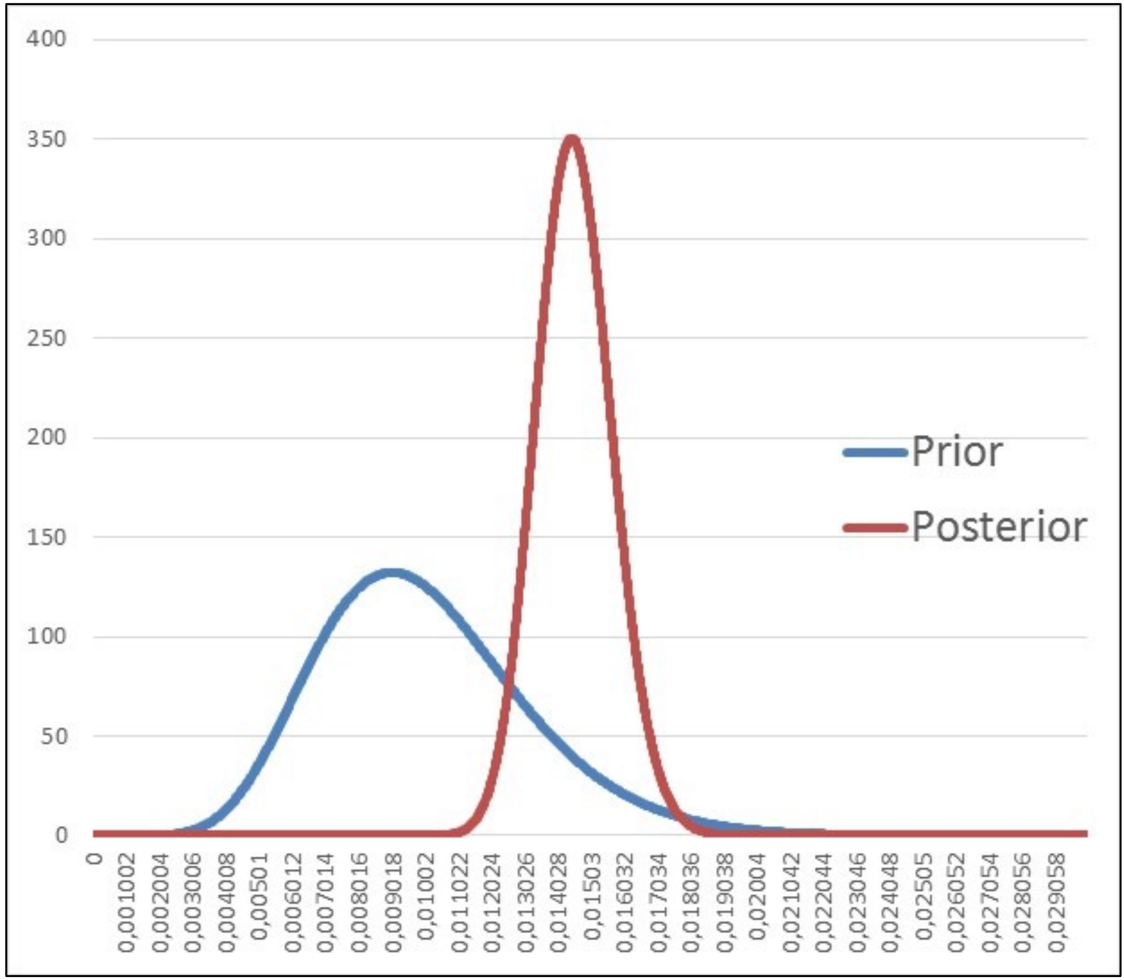
Example:

- Prior distribution: Beta(*10*, *990*): $\mathbb{E}_{prior} = 1\%$ CTR
- Data: 10,000 impressions with 150 clicks (1.5% CTR)

➢ Posterior distribution: Beta(*10+150*, *990+10000-150*):

$\mathbb{E}_{posterior} \sim 1.5\%$

	Clicks	No-Clicks	Expected CTR
Prior	10	990	1.0%
Data	150	9850	1.5%
Posterior	160	10840	1.5%



Choosing a Prior Distribution: Beta Distribution

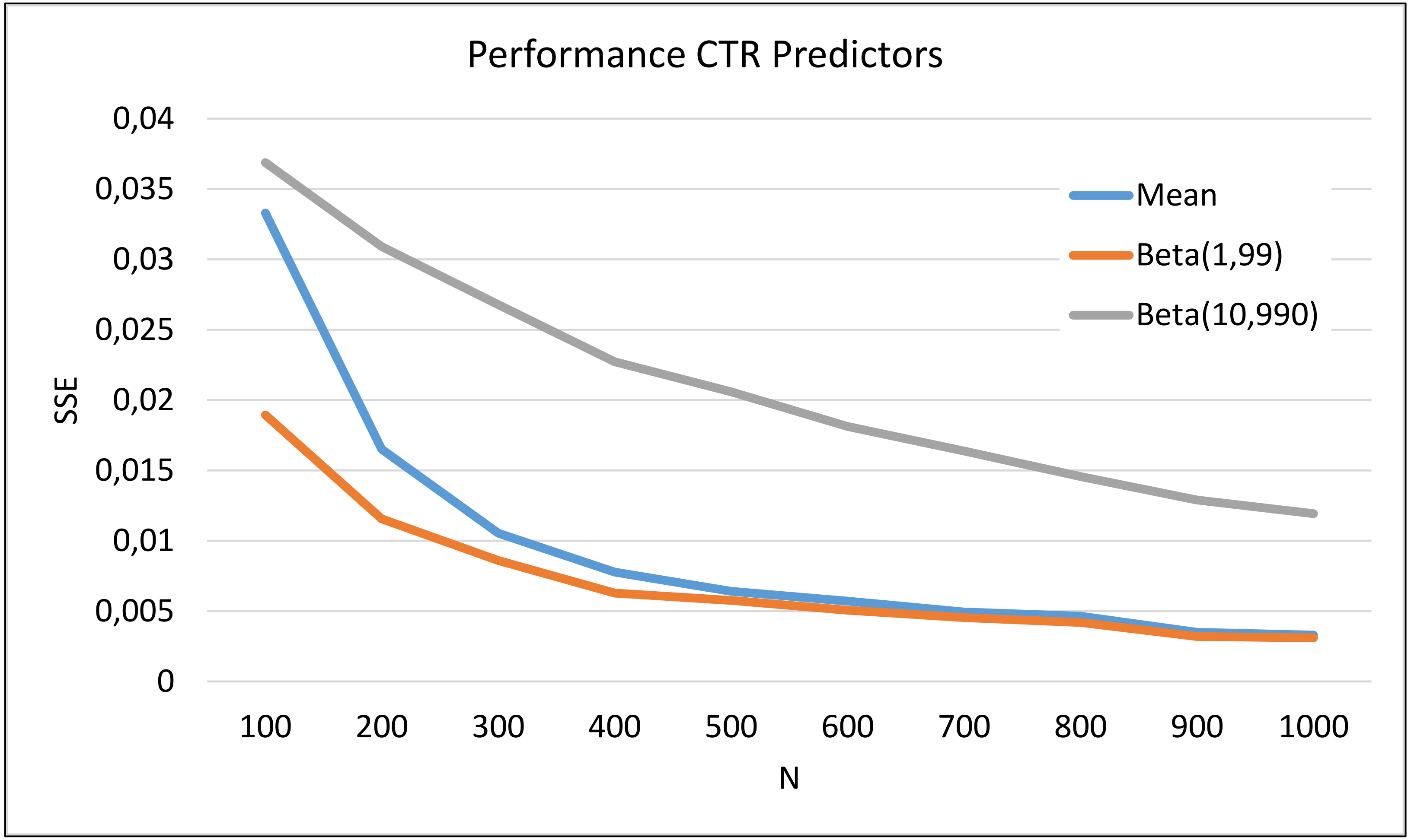
About Beta Distribution:

- Beta distribution is very flexible
- Beta distribution is the conjugate prior distribution of Bernoulli, Binomial, Negative Binomial and Geometric. This means that if you start Bayesian Updating with these prior distributions, eventually Beta comes out.

Bayesian Updating – Simulation Example

- Suppose the actual CTR is 0.0033
- We consider 3 CTR estimators:
    - Sample mean
    - Bayesian Updating with prior Beta(1, 99)
    - Bayesian Updating with prior Beta(10, 990)
- Note that both Beta priors have an initial CTR prediction equal to 0.01 which is too high
- Simulate clicks and see how fast our estimators converge to the actual CTR
- Evaluate the estimators for the following number of observations $N \in \{100, 200, \ldots, 1000\}$
- 1000 simulations for each N
- Evaluation based on the MSE of the predicted vs the actual CTR

Bayesian Updating – Simulation Results



Bayesian Updating – Conclusions

- Bayesian updating can improve estimates in start-up phase
- But performance depends greatly on the chosen prior
- Good prior gives:
  - Initial prediction close to actual (even when there is no data)
  - Fast learning/convergence
- Bad prior gives:
  - Bias in initial prediction
  - Slow learning/convergence

- Next week more advanced CTR prediction models

Conclusions

- In e-commerce we typically face **very small click-though rates** (CTR)
- **Many observations required for a reliable CTR estimate**
- **Bayesian Updating** can result in a more reliable CTR estimate relevatively fast
- The sooner we have good insights, the sooner we can **make decisions and spend our advertising money effectively**

Question: Can we <u>use what we already know</u> while <u>obtaining reliable estimates</u>?

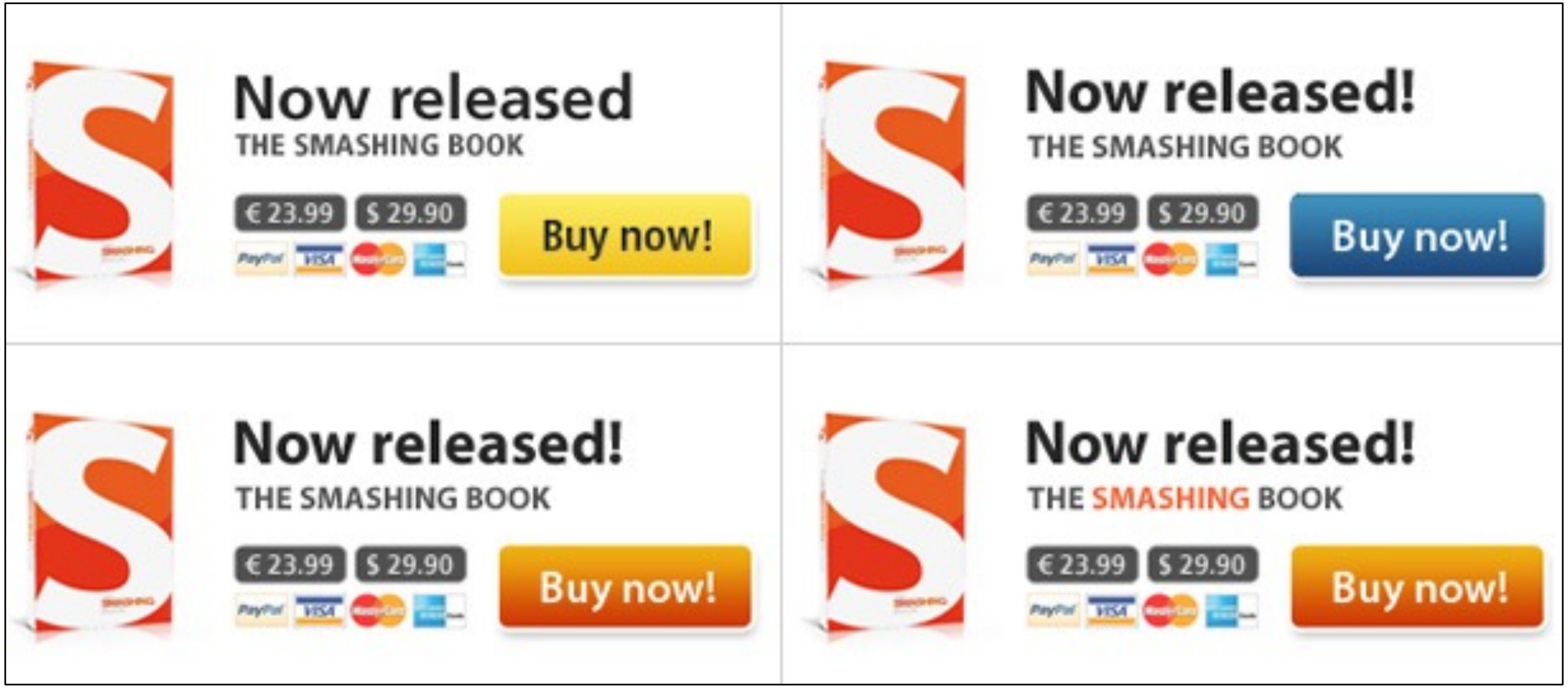
exploit

explore

Exploration vs Exploitation

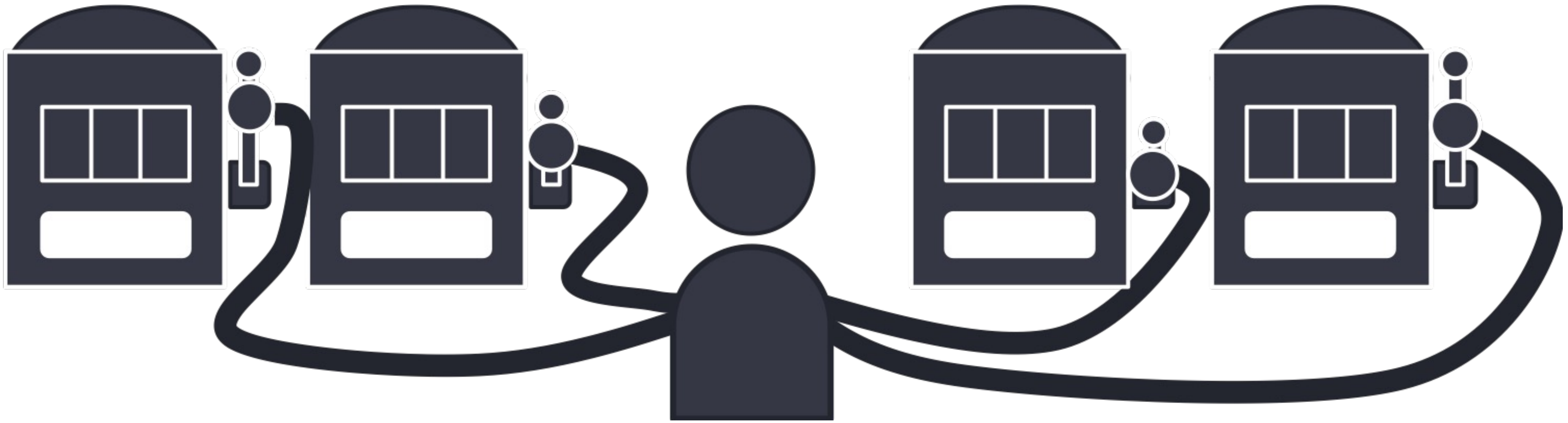
What version do we show next time?

CTR = 0.0009 CTR = 0.0011



CTR = 0.0012 CTR = 0.0015

Multi-Armed Bandit Problem



Multi-Armed Bandit - Methods

- A/B testing
- ε-Greedy
- Upper Confidence Bound (UCB)
- Thompson Sampling

A/B Testing

- Play each arm N times (explore) to find out which is the best one
- After that play only the best one (highest mean sample reward) (exploit)

About A/B Testing

Pros:

- Simple

Cons:

- Fixed time spent on exploring (often too much) before exploiting.
- Performance depends on chosen value for N
- No change in strategy once decided: you stop learning! What if you chose the wrong one or things change?

ε-Greedy

For each round:

- For each arm: estimate the expected reward (e.g., using Bayesian Updating)
- With probability 1-ε: play the arm with the highest expected reward
- With probability ε: play another arm, chosen randomly

ε-Greedy – Example

- Current estimates:

Beta(1, 1024), E = 0.0010

Beta(1, 1099), E = 0.0009

Beta(2, 1999), E = 0.0010

Beta(7, 4999), E = 0.0014

ε-Greedy – Example

- Current estimates:
- Next Impression (*ε=5%*):
  - With probability 95%: Show the ad with the highest estimated CTR (orange text, orange button)
  - With probability 5%: Show another ad randomly
  - Update the estimate of the shown ad based on the observation (click/no-click)

Beta(1, 1024), E = 0.0010

Beta(1, 1099), E = 0.0009

Beta(2, 1999), E = 0.0010

Beta(7, 4999), **E = 0.0014**

About ε-Greedy

Pros:

- We use the information that becomes available right away
- We don't get stuck in a sub-optimal state indefinitely

Cons:

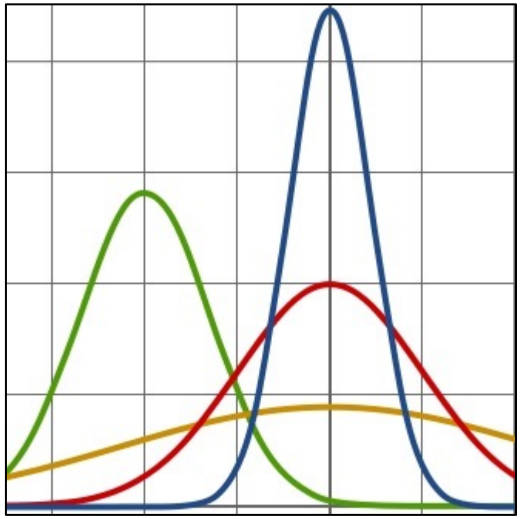
- Performance depends on chosen value for ε
- It can take a long time to get out of the current state
- If we are pretty sure, we still waste ε% of our budget on exploring

Upper Confidence Bound (UCB)

For each round:

- For each arm: calculate the α^{th} percentile Upper Confidence Bound (UCB) from the estimated reward distrbution (e.g., using Bayesian Updating)

- Play the arm with the highest UCB value



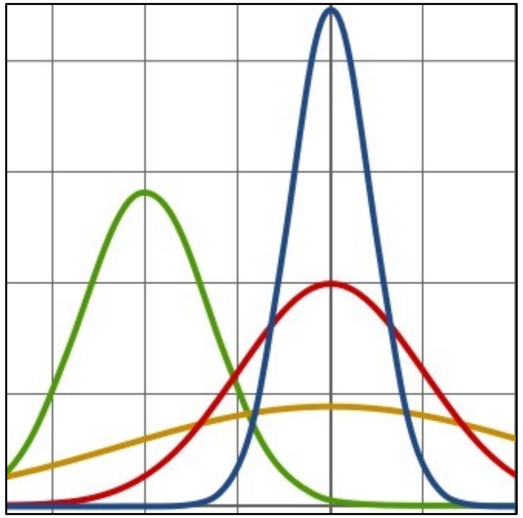
Upper Confidence Bound (UCB)

"Optimism in the face of uncertainty"

- Distributions have a longer tail when there is more uncertainty, and become more tight around the mean as we learn more

- The method has proven bounds for its worst case performance

- Popular values for α:
  - $\alpha = 1 - 1/N$
  - $\alpha = 95\%$ confidence interval (from Normal approximation)



UCB – Example

- Current estimates:

Beta(1, 1024), E = 0.0010

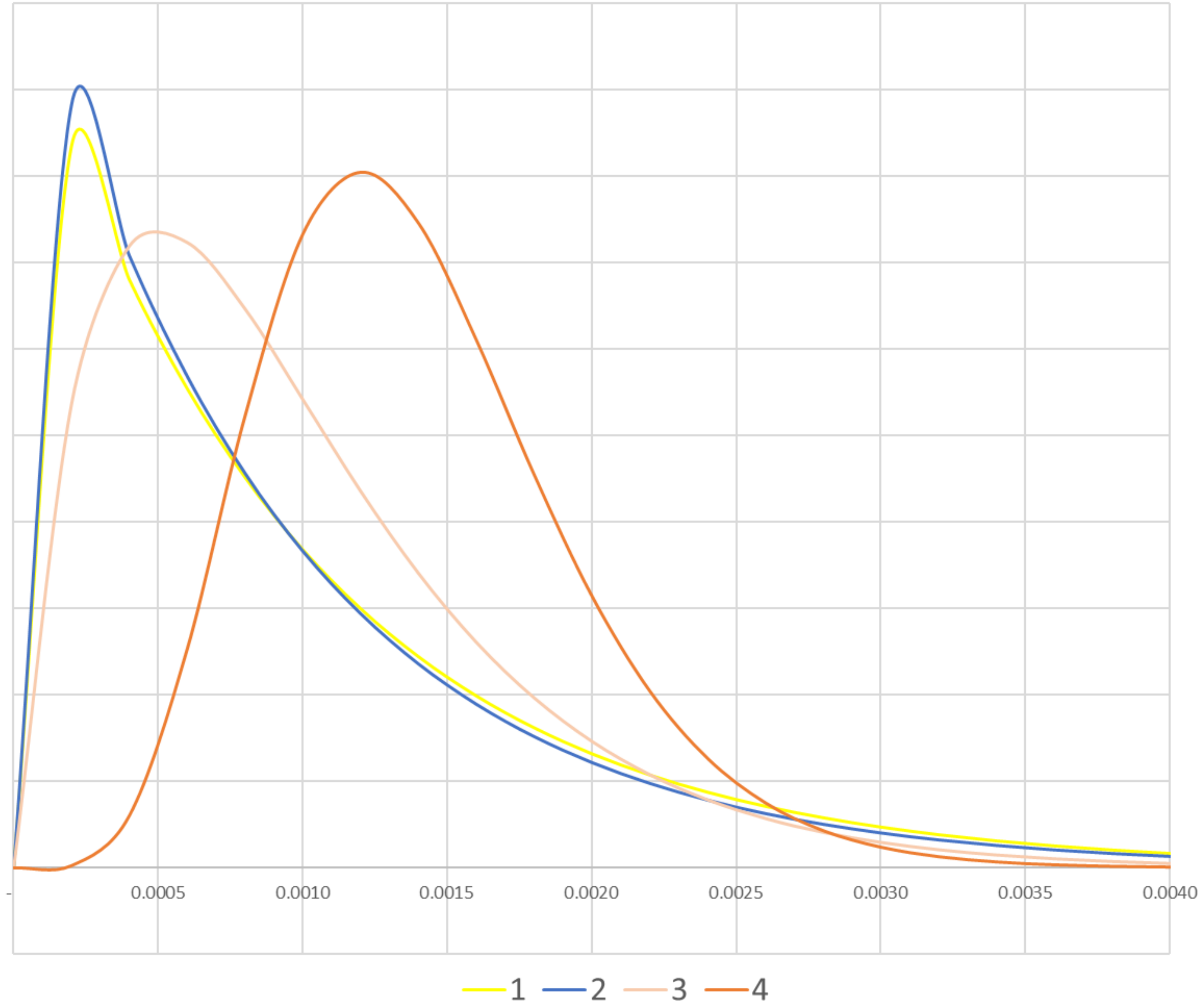
Beta(1, 1099), E = 0.0009

Beta(2, 1999), E = 0.0010

Beta(7, 4999), E = 0.0014

UCB – Example

- Distributions:



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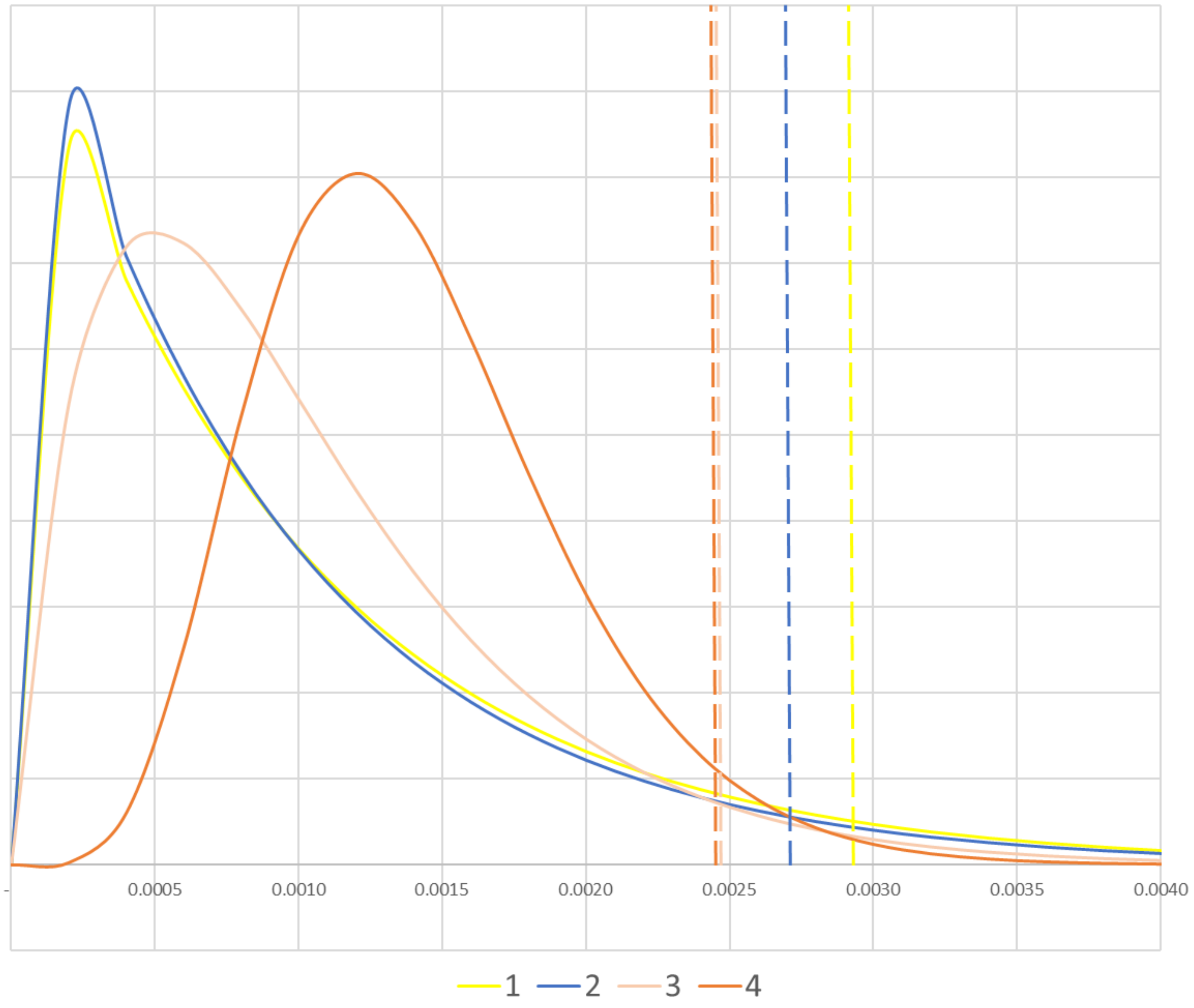
UCB – Example

- Distributions and 95% UCB:



UCB – Example

- Current estimates:
- Next Impression:
  - Show the ad with the highest UCB (black text, yellow button)
  - Update the estimate of the shown ad based on the observation (click/no-click)

Beta(1, 1024), **UCB = 0.0029**

Beta(1, 1099), UCB = 0.0027

Beta(2, 1999), E = 0.0024

Beta(7, 4999), UCB = 0.0024

About UCB

Pros:

- We use the information that becomes available right away
- We don't get stuck in a sub-optimal state indefinitely
- Balances exploring vs. exploiting
- Proven bounds to worst case performance

Cons:

- Performance depends on chosen value for α

Thompson Sampling

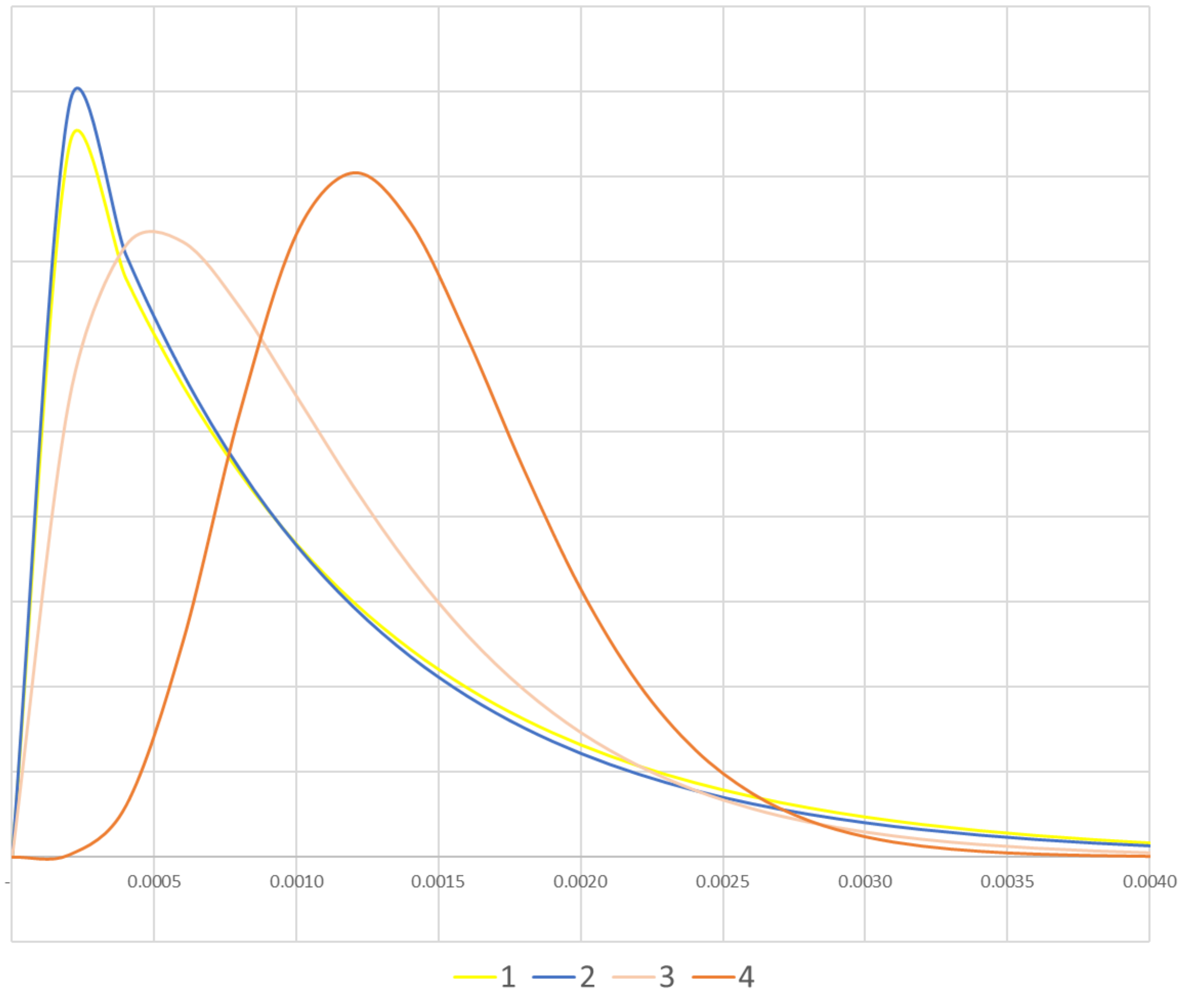
For each round:

- For each arm: draw a random value from the estimated reward distribution (using Bayesian Updating with a prior)
- Play the arm with the highest drawn value

Note: Updating the reward distribution does not need to be done in each round. This can be done after every *N* rounds. The randomness of the draw prevents us from playing the same arm every time again in between updates. This is a very powerful property for practical application.

Thompson Sampling – Example

- Distributions:



About Thomson Sampling

Pros:

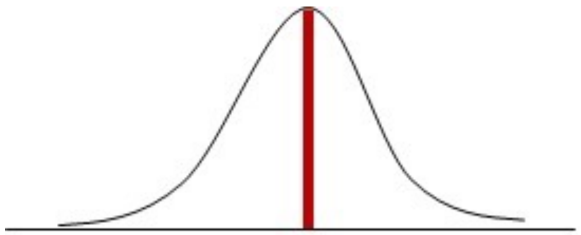
- We use the information that becomes available right away
- We don’t get stuck in a sub-optimal state indefinitely
- Uses the full reward function, not only the mean or UCB
- Updating the reward distribution is not needed after every round

Cons:

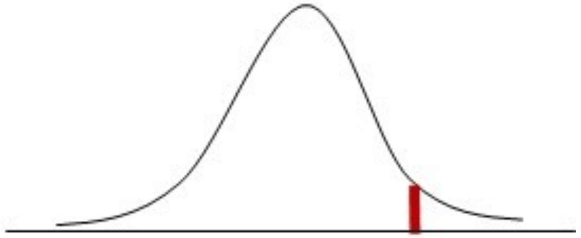
- Performance depends on chosen prior distributions

Multi Armed Bandit Solutions

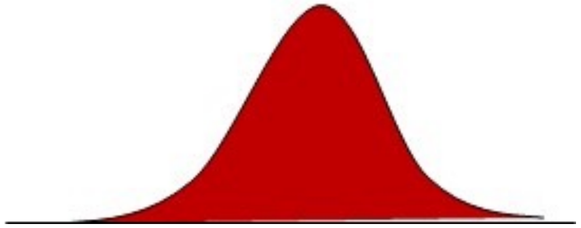
- A/B Testing
- ε-Greedy



- Upper Confidence Bound (UCB)



- Thompson Sampling



Example Simulation Results

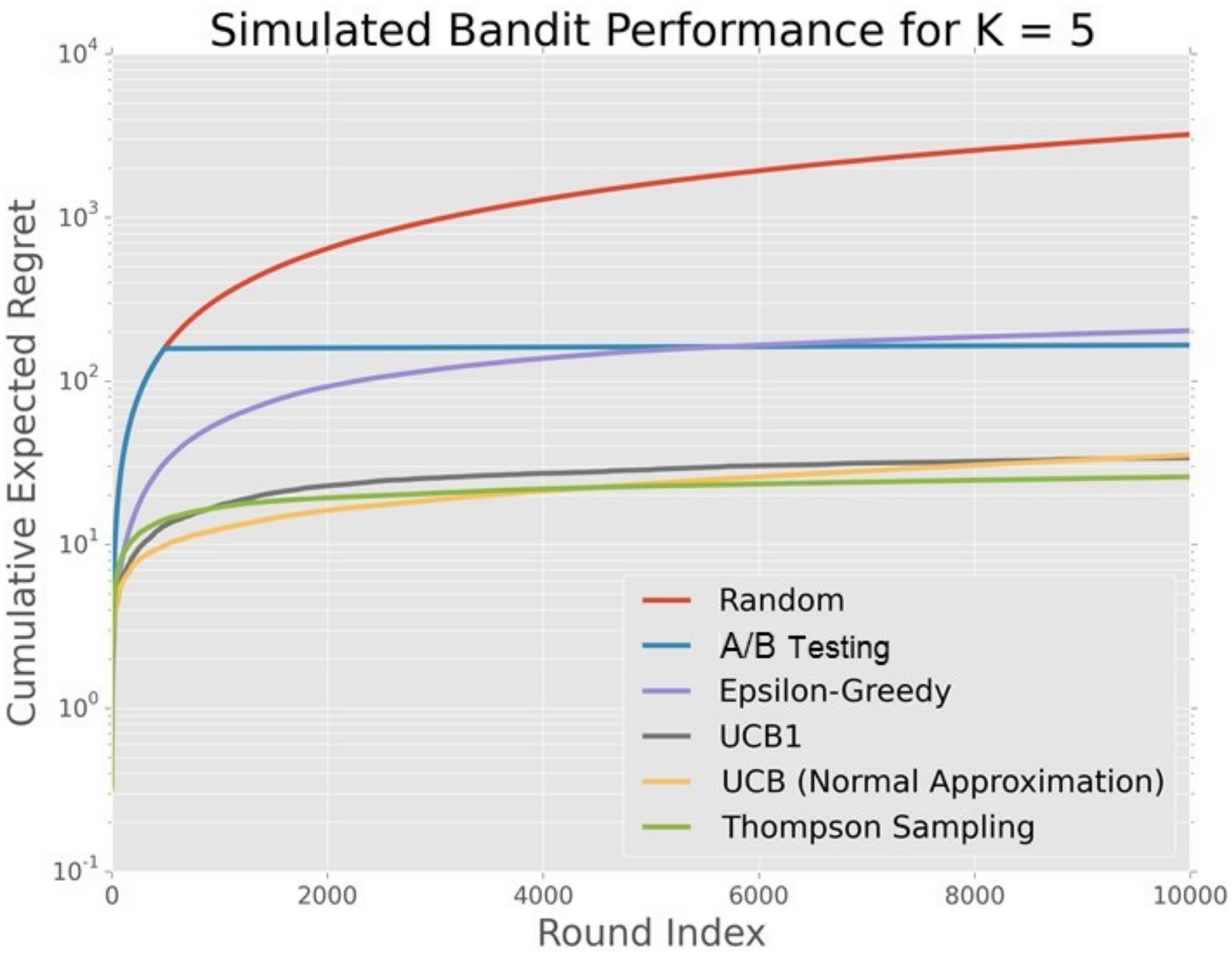


Image taken from: Recommendations with Thompson Sampling (Part II), S. Feldman, engineering.richrelevance.com, May 2014

Applications

Uber Eats - Mail Subject Line Optimization

Projects | Uber

MAB Results (Part II)

Performance of the treatment groups in terms of open rate (%)

Best Groups (Top 3)

Subject Line 8 (31.8%): Delicious food delivered to you
Subject Line 5 (31.5%): Ready to order, [First name]?
Subject Line 7 (31.3%): Prepare to feast, [First name]?

Worst Groups (Bottom 3)

Subject Line 6 (28.7%): Delicious food awaits you, [First name]
Subject Line 3 (29.9%): Thanks for joining—order today!
Subject Line 2 (30.0%): Thank you for joining UberEATS, [First name]

Image taken from: Improving Business Campaigns Through Continuous Experiments, Jeremy Gu, Uber Technologies Inc., AI Expo North America 2018, Santa Clara.

DoorDash – Cuisine Filters

Image taken from: Personalized Cuisine Filter, DoorDash Engineering blog, January 27, 2020

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Netflix - Image Selection

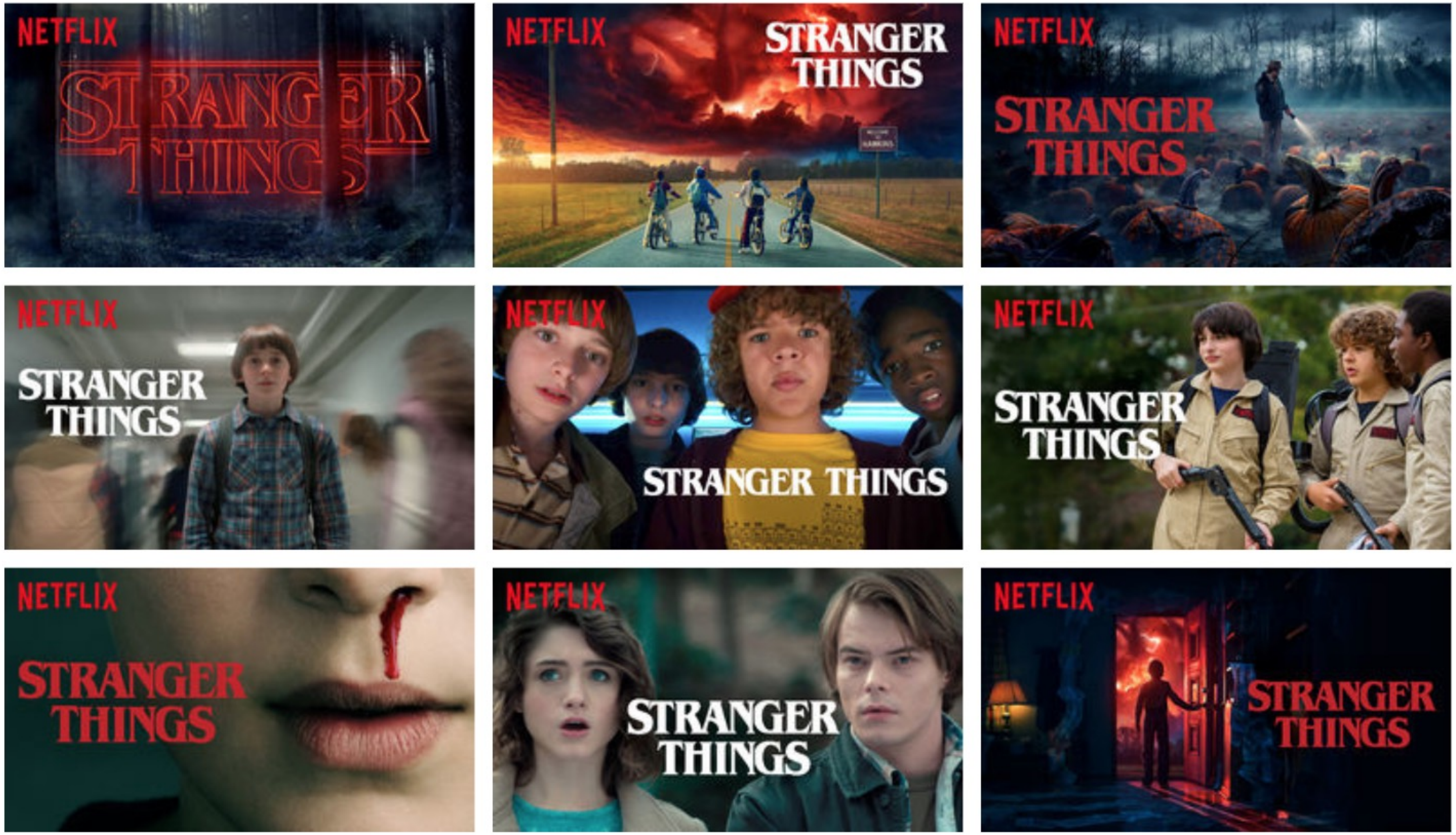

Image taken from: Artwork Personalization at Netflix; Chandrashekar, Amat, Basilico Jebara; https://netflixtechblog.com/artwork-personalization-c589f074ad76; Dec 7, 2017

Today's Course Material

- Lecture Notes
- Exam style exercises